```
data <- USArrests

cat("Summary Statistics:\n")

print(summary(data))

# Largest rape rate

max_rape <- which.max(data$Rape)

cat("\nState with largest Rape arrests:\n")

print(rownames(data)[max_rape])

# Maximum and minimum Murder

cat("\nState with maximum Murder rate:\n")

print(rownames(data)[which.max(data$Murder)])

cat("\nState with minimum Murder rate:\n")

print(rownames(data)[which.min(data$Murder)])

# Correlation

cat("\nCorrelation Matrix:\n")

print(cor(data))

# Above median Assault

median_assault <- median(data$Assault)

above_median <- data[

  data$Assault > median_assault,

]

cat("\nStates above median Assault:\n")

print(above_median)

# Bottom 25% Murder

q25 <- quantile(data$Murder, 0.25)

bottom_murder <- data[

  data$Murder <= q25,
```

```
]

cat("\nBottom 25% Murder states:\n")

print(bottom_murder)

# Histogram

hist(

  data$Murder,

  main = "Murder Arrests",

  xlab = "Murder"

)

# Density

plot(

  density(data$Murder),

  main = "Density of Murder Arrests",

  xlab = "Murder"

)

# Scatter plot

plot(

  data$Murder,

  data$Assault,

  main = "Murder vs Assault",

  xlab = "Murder",

  ylab = "Assault"

)

# Bar graph

barplot(

  data$Murder[1:10],
```

```
  names.arg = rownames(data)[1:10],

  las = 2,

  main = "Murder Rates - First 10 States"

)
```